from the beginnings of the century, the second from the 1930s, and the last two from the mid-1950s—were little more than scientific romances, hardly constrained by the sparse data available.

Then, in 1956, a report was published in *The Astrophysical Journal* by Cornell H. Mayer and his colleagues. They had pointed a newly completed radio telescope, built in part for classified research, on the roof of the Naval Research Laboratory in Washington, D.C., at Venus and measured the flux of radio waves arriving at Earth. This was not radar: No radio waves were bounced off Venus. This was listening to radio waves that Venus on its own emits to space. Venus turned out to be much brighter than the background of distant stars and galaxies. This in itself was not very surprising. Every object warmer than absolute zero (–273˚C) gives off radiation throughout the electromagnetic spectrum, including the radio region. You, for example, emit radio waves at an effective or "brightness" temperature of about 35˚C, and if you were in surroundings colder than you are, a sensitive radio telescope could detect the faint radio waves you are transmitting in all directions. Each of us is a source of cold static.

What *was* surprising about Mayer's discovery was that the brightness temperature of Venus is more than 300˚C, far higher than the surface temperature of the Earth or the measured infrared temperature of the clouds of Venus. Some places on Venus seemed at least 200˚ hotter than the normal boiling point of water. What could this mean?

Soon there was a deluge of explanations. I argued that the high radio brightness temperature was a direct indication of a hot surface, and that the high temperatures were due to a massive carbon dioxide/water vapor greenhouse effect—in which some sunlight is transmitted through the clouds and heats the surface, but the surface experiences enormous difficulty in radiating back to space because of the high infrared opacity of carbon dioxide and water vapor. Carbon dioxide absorbs at a range of wavelengths through the infrared, but there seemed to be "windows" between the CO_2 absorption bands through which the surface could readily